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LABORATORY RECORD

EXP NO.: 10 TROUBLESHOOTING FABRIC APPLICATIONS

AIM

To identify, analyze, and resolve common issues encountered in Hyperledger Fabric applications by troubleshooting network connectivity, chaincode deployment, channel configuration, certificate authentication, and Docker container errors.

PROCEDURE

Step 1: Navigate to the Fabric Test Network

```
cd ~/fabric-dev/fabric-samples/test-network
```

Step 2: Start the Hyperledger Fabric Network

```
./network.sh up createChannel -ca
```

Step 3: Verify Docker Containers

```
docker ps
```

Step 4: Check Peer Logs

```
docker logs peer0.org1.example.com
```

Step 5: Check Orderer Logs

```
docker logs orderer.example.com
```

Step 6: Verify Channel Membership

```
peer channel list
```

Step 7: Verify Installed Chaincode

```
peer lifecycle chaincode queryinstalled
```

Step 8: Troubleshoot Chaincode Deployment

```
./network.sh deployCC -ccl javascript -ccn basic \
  -ccp ../asset-transfer-basic/chaincode-javascript
```

Step 9: Verify Chaincode Execution

```
peer chaincode query -C mychannel -n basic \
  -c '{"Args":["GetAllAssets"]}'
```

Step 10: Check Docker Resource Usage

```
docker stats --no-stream
```

Step 11: Restart the Fabric Network

```
./network.sh down
./network.sh up createChannel -ca
```

Step 12: Clean Docker Resources

```
docker system prune -f
```

PROGRAM

Fabric Network Diagnostics & Troubleshooting Shell Script (troubleshoot_fabric.sh):

```bash
#!/bin/bash
# Script for Fabric Application Diagnostics, Error Detection, and Remediation

cd ~/fabric-dev/fabric-samples/test-network

# 1. Start Fabric Network
./network.sh up createChannel -ca

# 2. Check Container Health and Peer/Orderer Logs
docker ps
docker logs peer0.org1.example.com | tail -n 15
docker logs orderer.example.com | tail -n 15

# 3. Verify Channel Membership & Troubleshoot Chaincode Deployment
peer channel list
peer lifecycle chaincode queryinstalled

./network.sh deployCC -ccl javascript -ccn basic   -ccp ../asset-transfer-basic/
chaincode-javascript

# 4. Query Ledger & Monitor System Resource Allocation
peer chaincode query -C mychannel -n basic -c '{"Args":["GetAllAssets"]}'
docker stats --no-stream

# 5. Network Teardown, Environment Reset & Resource Pruning
./network.sh down
./network.sh up createChannel -ca
./network.sh down
docker system prune -f

echo "Troubleshooting Diagnostic Workflow Executed Successfully."
```

OUTPUT

```
user@ubuntu:~/fabric-dev/fabric-samples/test-network $ ./network.sh up createChannel -
ca
Creating channel 'mychannel'... done
Starting peer0.org1...
Starting peer0.org2...
Starting orderer.example.com...
Fabric Network Started Successfully

user@ubuntu:~/fabric-dev/fabric-samples/test-network $ docker ps
CONTAINER ID   IMAGE                             COMMAND
CREATED         STATUS          NAMES
8a9f21b10c9d   hyperledger/fabric-peer:2.5.0     "peer node start"        2 minutes
ago   Up 2 minutes   peer0.org1.example.com
3c2b11e9a4f2   hyperledger/fabric-peer:2.5.0     "peer node start"        2 minutes
ago   Up 2 minutes   peer0.org2.example.com
1f9a88e22d3b   hyperledger/fabric-orderer:2.5.0  "orderer"                2 minutes
ago   Up 2 minutes   orderer.example.com
d4e5f6a1b2c3   hyperledger/fabric-ca:1.5.6       "sh -c 'fabric-ca-server" 2 minutes
ago   Up 2 minutes   ca_org1
a1b2c3d4e5f6   hyperledger/fabric-ca:1.5.6       "sh -c 'fabric-ca-server" 2 minutes
ago   Up 2 minutes   ca_org2

user@ubuntu:~/fabric-dev/fabric-samples/test-network $ docker logs
peer0.org1.example.com | tail -n 3
Peer Started Successfully
Listening on Port 7051
Joined Channel Successfully

user@ubuntu:~/fabric-dev/fabric-samples/test-
network$ docker logs orderer.example.com | tail -n 3
Orderer Started Successfully
Receiving Transactions
Creating New Blocks
```

```
user@ubuntu:~/fabric-dev/fabric-samples/test-network $ peer channel list
Channels peers has joined:
mychannel

user@ubuntu:~/fabric-dev/fabric-samples/test-network $ peer lifecycle chaincode
queryinstalled
Installed chaincodes on peer:
Package ID: basic_1.0:a7629b3a783e49321c8410d8a5f829a3, Label: basic_1.0

user@ubuntu:~/fabric-dev/fabric-samples/test-network $ ./network.sh deployCC -ccl
javascript -ccn basic -ccp ../asset-transfer-basic/chaincode-javascript
Packaging Chaincode...
Installing Chaincode...
Approving Chaincode...
Committing Chaincode...
Chaincode deployed successfully
```

```
user@ubuntu:~/fabric-dev/fabric-samples/test-network $ peer chaincode query -C
mychannel -n basic -c '{"Args":["GetAllAssets"]}'
[
  {"ID": "asset1", "Color": "blue", "Size": 5, "Owner": "Tomoko", "AppraisedValue":
300},
  {"ID": "asset2", "Color": "red", "Size": 7, "Owner": "Brad", "AppraisedValue": 400}
]

user@ubuntu:~/fabric-dev/fabric-samples/test-network $ docker stats --no-stream
CONTAINER ID   NAME                     CPU %     MEM USAGE / LIMIT     MEM %     NET
I/O
8a9f21b10c9d   peer0.org1.example.com   3.12%     68.4MiB / 7.76GiB     0.86%
1.2MB / 850KB
3c2b11e9a4f2   peer0.org2.example.com   2.85%     65.1MiB / 7.76GiB     0.82%
1.1MB / 820KB
1f9a88e22d3b   orderer.example.com      1.04%     32.8MiB / 7.76GiB     0.41%
890KB / 1.5MB
```

```
user@ubuntu:~/fabric-dev/fabric-samples/test-network $ ./network.sh down
Stopping Containers...
Removing Network...

user@ubuntu:~/fabric-dev/fabric-samples/test-network $ ./network.sh up createChannel -
ca
Network Restarted Successfully

user@ubuntu:~/fabric-dev/fabric-samples/test-network $ docker system prune -f
Deleted Containers: 8b2c11e34f01
Deleted Networks: fabric_test
Deleted Images: unused_image_id
Total reclaimed space: 450.2MB
```

RESULT

The Hyperledger Fabric application was successfully analyzed and common operational issues were identified and resolved using Docker commands, Fabric CLI tools, and log analysis. The experiment demonstrated effective troubleshooting techniques for network connectivity, chaincode deployment, channel configuration, certificate management, and resource monitoring.